



GC1015

CONDITIONAL RISK, EXPECTED VALUE, AND SIMULATION

Session 4



Use what you already know

RECALL

Two questions from Session 3. Answer from memory: no notes, no slides.

1

From Session 3: measurement scales

- Why can an arithmetic mean of satisfaction tiers coded 1 to 5 mislead?

2

From Session 3: describing distributions

- In heavily right-skewed data, why does the mean stop describing a typical case?

3

How to answer

- Write one sentence for each question in 2 minutes.
- Mark how sure you are: 1 = guess, 2 = fairly sure, 3 = certain. The next two slides check your answers.

CHECK YOUR RECALL (1 OF 2)

Scene: *You are CloudFlow's analyst. The customer team scored a survey from 1 (very unhappy) to 5 (very happy) and wants one average score on the committee's first slide.*

Why can an arithmetic mean of satisfaction tiers coded 1 to 5 mislead?

- ☐ A A mean assumes equal gaps between tiers; ordinal tiers guarantee only their order.
- ☐ B Survey samples for each tier are always too small to average.
- ☐ C Survey categories must stay as words; coding them as numbers breaks the rules.

Choose, then explain your reason to a neighbor. Compare it with the sentence you wrote.

Why it matters today: in Section 1 you count CloudFlow accounts by category (Churn: Yes or No; HighTickets: Yes or No). Categories are counted and turned into proportions, not averaged as codes.

CHECK YOUR RECALL (2 OF 2)

Scene: You are CloudFlow's analyst. Most support tickets close within a day; some stay open for weeks. The ops lead wants one "typical" fix time for the board.

In heavily right-skewed data, why does the mean stop describing a typical case?

- A** Extreme high values drag the mean up toward the tail, away from most cases.
- B** Right skew stops the standard deviation from being calculated properly.
- C** In right-skewed data, the median automatically drops to zero.

Choose, then explain your reason to a neighbor. Compare it with the sentence you wrote.

Why it matters today: one average can mislead in another way. In Section 3, a capacity plan built on average demand overstates CloudFlow's average profit.



*Four steps from risk
signals to capital decisions*

TODAY'S INVESTIGATION

Four steps from risk signals to capital decisions

1

Before the break

- **1. Probability rules and conditional risk:** update churn risk after a high-ticket flag.
- **2. Expected monetary value:** compare capacity choices across uncertain demand.

2

After the break

- **3. Monte Carlo simulation:** model profit across 1,000 trials and see the flaw of averages.
- **4. Group studio:** advise CloudFlow's committee with your lab evidence, in the PNP-GRP.



*Slides hand off to the lab,
the lab hands back*

TODAY'S LOOP: SLIDES AND LAB

Every section ends in the lab; every practice returns you here

1

Sections before the break

- Each section reserves time for an SPSS demo. Then complete the timed Excel practice, evidence caption and interpretation.

2

Break after Section 2

- Return for Sections 3 and 4.

3

Then the studio and the close

- Group studio (PNP-GRP: Capital Investment Risk Governance), then synthesis.
Open now: the LAB02 guide, your LAB02 workbook in Excel and **LAB02_SPSS.sps**.

WHY THIS SESSION MATTERS



Key point

A familiar smoke alarm illustrates conditional risk: an alarm is evidence of fire, but it does not prove that a fire exists.



Expected value

is a probability-weighted average across possible outcomes, not a guaranteed result.



Simulation

repeatedly samples uncertain inputs to reveal the distribution, variability, and downside that a single average can hide. These tools help managers distinguish a useful signal from certainty and a promising average from an acceptable risk.

THE CLOUDFLOW CASE: WHAT THE CFO NEEDS

From the Chief Financial Officer and Executive Risk Committee to you, the Senior Business Risk Analyst

Question for today	CloudFlow evidence	Where
Which accounts carry higher churn risk, and can a high-ticket flag guide an automated screening rule?	1,000 recorded accounts in cloudflow_churn_cohort.csv: 120 churned, 272 high-ticket, 96 both	Section 1 · Lab Part A
Which capacity alternative has the higher expected value?	Large Hub pays \$12M, \$3M or -\$9M; Modular Hub \$9M, \$5M or \$1M, for high, medium or low demand (30%, 50%, 20%)	Section 2 · Lab Part B
What happens to profit when demand is uncertain?	Demand averages 1,000 units (SD 250). Capacity is 1,000 units at \$40 each; price \$100; unused units salvage at \$15	Section 3 · Lab Part C
Which three-year capital option balances return against the risk of a loss?	Option A (\$1.0M) and Option B (\$0.9M) initial investment; three years at an 8% discount rate	Take-home (after class)

Two limits from the CFO: a flag that goes with churn does not prove that screening will prevent churn, and an expected value is not a guaranteed payoff.

Section 4 is the group studio: your group uses rows 1 to 3 to advise the committee | Source: LAB02 guide, stakeholder request and data dictionary

DECISION POINT: SCREENING AND CAPACITY

Advise the committee now; your group decides in Section 4.



Situation

CloudFlow's records show that high-ticket accounts churn more often than others. Leaders want to screen those accounts and fund new capacity now, while demand is uncertain.

A

Commit and reach out

Reserve capacity now; call every high-ticket account. Benefit: fast fixes. Risk: wasted calls to loyal accounts.

B

Hold and review

Keep capacity steady; managers review flags first. Benefit: capital stays flexible. Risk: slower help loses accounts.

Your decision

Scene: *Friday, 5:40 p.m. The committee chair stops you at the elevator and asks for your first choice before any numbers arrive.*

Which would you advise today? Your group gives the final answer in Section 4.

SECTION 1

PROBABILITY RULES AND CONDITIONAL BUSINESS RISK

Compute and interpret marginal, joint and conditional probabilities from CloudFlow's account table, then use Bayes rule to update churn risk.

In this section: contingency tables · conditional probability and Bayes rule



*Probability Essentials and
Contingency Tables*

ANATOMY OF TWO-WAY CONTINGENCY TABLES

Where single events, overlaps and totals sit in a table

1

Inside cells and margins

- A two-way table sorts every account by two categories, such as **Churn** and **HighTickets**. Inside cells count accounts that meet both (joint). Row and column totals count one at a time (marginal).

2

Addition rule

- Either event, or both: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. Subtract the overlap once so accounts in both groups are not counted twice.

3

Try it with CloudFlow's table

- **Inside cell:** accounts that are high-ticket and churned. **Margin:** all churned accounts, whatever their tickets.
- **Check:** the four inside cells add up to the grand total, and each row total equals its two cells.

CALCULATING MARGINAL AND JOINT PROBABILITIES

1,000 recorded CloudFlow accounts by HighTickets and Churn, laid out like lab Part A (B9:D11)

HighTickets \ Churn	Churn = Yes	Churn = No	Total
HighTickets = Yes	96	176	272
HighTickets = No	24	704	728
Total	120	880	1,000

Marginal: $P(\text{Churn}) = 120 \div 1,000 = 0.120$ and $P(\text{HighTickets}) = 272 \div 1,000 = 0.272$. Joint: $P(\text{Churn and HighTickets}) = 96 \div 1,000 = 0.096$.

One row per recorded account in cloudflow_churn_cohort.csv. Churn and HighTickets are Yes or No; a blank is missing, never No | Unit: accounts

GENERAL ADDITION RULE OF PROBABILITY

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

In words: add the two groups, then remove the overlap once, so no account is counted twice.

Symbols

$P(A \text{ or } B)$, also written $P(A \cup B)$: A, B or both happen

$P(A)$, $P(B)$: marginal probabilities

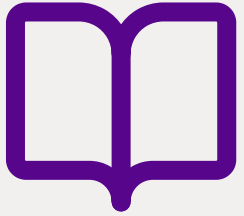
$P(A \text{ and } B)$, also written $P(A \cap B)$: both happen (joint)

CloudFlow, from the table

$$0.120 + 0.272 - 0.096 = 0.296$$

Result: 29.6% of accounts churned, were high-ticket, or both.

Limit: adding without subtracting works only when the two events cannot happen together (mutually exclusive).



*Conditional Probability
and Bayesian Updating*

BAYES RULE: HOW A FLAG UPDATES RISK

Conditional probability, total probability and Bayes rule, step by step

1

Conditional probability

- Look only inside the condition: $P(A | B) = P(A \text{ and } B) \div P(B)$. Read the bar as "given".

2

Law of total probability

- A flag can come from churners or from renewers, so add both paths: $P(B) = P(B | A)P(A) + P(B | \text{not } A)P(\text{not } A)$.

3

Bayes rule

- Posterior = the churn path $\div$ all flagged accounts. It updates the prior, the baseline risk before the flag.

BAYES RULE WITH CLOUDFLOW'S NUMBERS

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B|A) \cdot P(A) + P(B|A^c) \cdot P(A^c)}$$

Read A^c as "not A". Here A = Churn, so A^c = Renew, and B = the high-ticket flag.

Inputs

Prior: $P(\text{Churn}) = 0.12$, so $P(\text{Renew}) = 0.88$

Flag among churners: $P(\text{HighTickets} | \text{Churn}) = 0.80$

Flag among renewers: $P(\text{HighTickets} | \text{Renew}) = 0.20$

Steps

1. Churn path: $0.80 \times 0.12 = 0.096$
2. Renew path: $0.20 \times 0.88 = 0.176$
3. All flagged: $0.096 + 0.176 = 0.272$
4. Posterior: $0.096 \div 0.272 = 0.353$

Result: a high-ticket account has a 35.3% chance of churning, up from 12%.

WORKED EXAMPLE: UPDATING CHURN RISK FOR CLOUDFLOW

The same numbers as a table: out of 1,000 accounts, who gets flagged?

Account status	Prior	P(HighTickets status)	Joint (flagged share)	Posterior	Accounts flagged (of 1,000)
Churn	0.12	0.80	0.096	0.353	96
Renew	0.88	0.20	0.176	0.647	176
Total	1.00		0.272	1.000	272

A high-ticket flag raises churn risk from 12.0% to 35.3%, yet most flagged accounts renew: 176 false alarms against 96 true churners.

CloudFlow cohort, 1,000 recorded accounts (lab Part A) | Unit: probabilities; accounts in the last column

AI SEARCH & SHARE



*Discuss with a partner;
Artificial intelligence (AI)
support is optional.*

APPLY THE IDEA

3 minutes with a partner; AI support is optional

1

Partner question

- Under what condition would the posterior $P(A | B)$ equal the prior $P(A)$?
- Ask an AI tool if you like, then **check its answer** with CloudFlow's numbers: is $P(\text{Churn} | \text{HighTickets})$ equal to $P(\text{Churn})$? Share one sentence.

2

How to check an AI answer

- Ask it to state the rule it used, not just the conclusion.
- Test the rule on CloudFlow: compare $P(\text{Churn} | \text{HighTickets}) = 35.3\%$ with $P(\text{Churn}) = 12\%$.

DECISION POINT: DOES THE FLAG JUSTIFY SCREENING?

Use the evidence, explain your decision, and name one limit.



Decision question

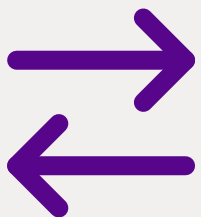
The flag catches 80% of churners. Why is a flagged account not 80% likely to churn, and should CloudFlow screen flagged accounts?

The evidence

96 of 272 flagged accounts churned: 35.3%, against a 12.0% baseline.
176 flagged accounts renewed (false alarms).

Name one limit

A flag that goes with churn does not show that screening will prevent churn.



Setup, then practice and explain

CLASS PRACTICE 1: CALCULATE CONDITIONAL CHURN RISK

8 minutes in the lab, then back to slide 20: Expected Monetary Value

1

Before the practice timer

- **Instructor demo (4 min):** follow along in SPSS block H1 and leave the output open.
- **Then:** open **IN-CLASS_Lab Part A** in your workbook and start the 8-minute timer.

2

Check and bring back

- **Check:** **B18** = 35.3%; the counts add to **D11**. **Bring back:** the denominator you used, and why it is not proof of cause.

3

If you need help, or finish early

- **If you need help:** **B18** divides the Yes/Yes count **B9** by the high-ticket total **D9**, not by the churn total **B11**.
- **If you finish early:** halve the false-alarm rate (20% of renewers → 10%) and recompute the posterior. Why does it jump?

SECTION 2

EXPECTED MONETARY VALUE AND DECISION ANALYSIS

Structure choices across uncertain demand with payoff tables, decision trees and the Expected Monetary Value criterion.

In this section: payoff tables and EMV · decision trees and risk attitude

Buffett's Handoff Puts \$365 Billion Behind New Decision Makers

WHAT HAPPENED

Associated Press reported that Warren Buffett, 96, became Berkshire Hathaway chairman emeritus immediately; Howard Buffett became chairman, while Greg Abel remained chief executive. The roughly \$1 trillion U.S. conglomerate held about \$365 billion in cash, making future investment and capital-allocation decisions unusually consequential.

The handoff puts a vast cash reserve under new leadership, making uncertain market states and alternative capital-allocation payoffs central to future choices.

Scene: *You are CloudFlow's analyst. The CFO forwards this story with one line: "Same puzzle as our hub decision, just with more zeros."*

How should the new leaders compare holding cash, buying companies and buying back shares while future markets are uncertain?

- A** Weight each choice's payoffs across market states by their probabilities.
- B** Hold all the cash until the economic uncertainty has fully cleared.
- C** Pick the choice with the biggest payoff in the best market.

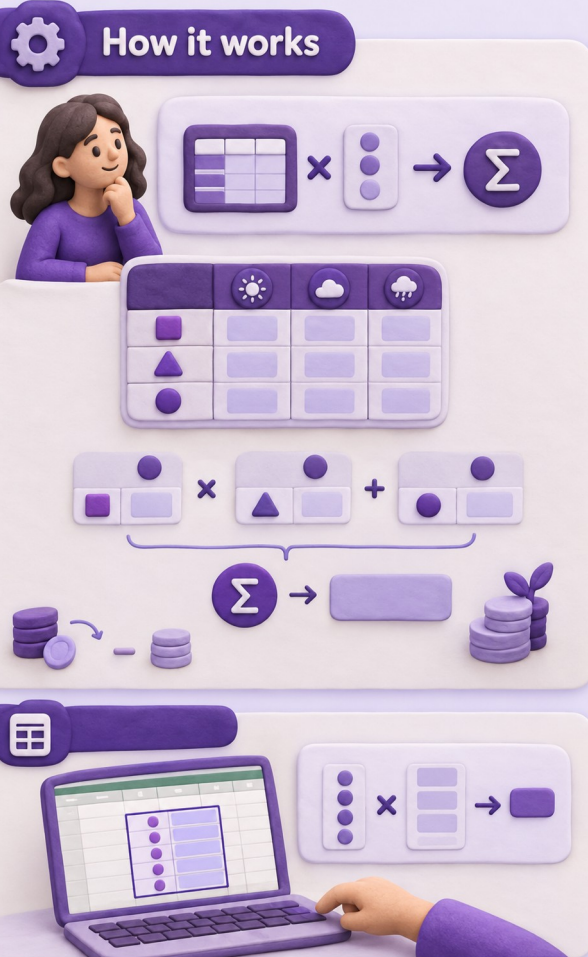


Reporter in the field covering Buffett's Handoff Puts \$365 Billion Behind New Decision Makers. [Illustrative image]

Source: Associated Press · 2026-09-18

MECHANISM OF CONSTRUCTING A PAYOFF TABLE AND CALCULATING EMV

Payoff Tables and Expected Monetary Value



1

How EMV is calculated

- Multiply each net payoff by its state probability, then add the products: $EMV = \text{SUM}(P_i \times \text{Payoff}_i)$. In Excel: **=SUMPRODUCT(probabilities, payoffs)**.

2

Where the numbers go

- Rows are the choices (Large Hub, Modular Hub). Columns are demand states with their probabilities. Each cell is a net payoff in \$ millions.

3

Two checks before you trust an EMV

- The probabilities cover every state once and add up to 100% (30% + 50% + 20%).
- EMV is a long-run average over many similar decisions, not the payoff CloudFlow will get this time.

EXPECTED MONETARY VALUE FORMULA (1/3)

$$EMV(d_j) = \sum_{i=1}^S P(s_i) \times v(d_j, s_i)$$

Symbols:

EMV(d_j): Expected Monetary Value of decision alternative d_j measured in currency units

P(s_i): Prior probability of state of nature s_i, where 0 ≤ P(s_i) ≤ 1 and sum of P(s_i) = 1.00

v(d_j, s_i): Net economic payoff or value realized when decision d_j is executed and state s_i occurs

S: Total number of mutually exclusive and collectively exhaustive states of nature

EXPECTED MONETARY VALUE FORMULA (2/3)

$$EMV(d_j) = \sum_{i=1}^S P(s_i) \times v(d_j, s_i)$$

Inputs: Probabilities $P(s_i) = [0.30, 0.50, 0.20]$ and Payoffs for Large Hub $v(\text{Large}, s_i) = [12.0, 3.0, -9.0]$ in millions of dollars.

Steps:

1. Multiply High Demand probability by High Demand payoff: $0.30 * 12.0 = 3.60$ million dollars.
2. Multiply Medium Demand probability by Medium Demand payoff: $0.50 * 3.0 = 1.50$ million dollars.
3. Multiply Low Demand probability by Low Demand payoff: $0.20 * (-9.0) = -1.80$ million dollars.

EXPECTED MONETARY VALUE FORMULA (3/3)

$$EMV(d_j) = \sum_{i=1}^S P(s_i) \times v(d_j, s_i)$$

Steps:

4. Sum all weighted products: $3.60 + 1.50 + (-1.80) = 3.30$ million dollars.

Result: $EMV(\text{Large Hub}) = 3.30$ million dollars

Over repeated identical market implementations, building the Large Hub yields an expected net economic value of 3.30 million dollars.

Use: Capital budgeting and investment appraisal across operational capacity projects.

Limit: Assumes a risk-neutral decision maker who can absorb a 9.0 million dollar loss without organizational failure.

WORKED EVIDENCE: CLOUDFLOW CAPACITY ALTERNATIVES

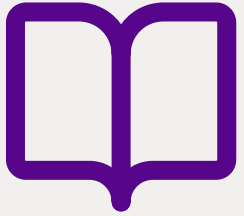
Payoff Tables and Expected Monetary Value

Alternative	High demand (30%)	Medium demand (50%)	Low demand (20%)	EMV
Large Hub	12.00	3.00	−9.00	3.30
Modular Hub	9.00	5.00	1.00	5.40

The Modular Hub has the higher EMV (\$5.40M against \$3.30M, a \$2.10M premium) and never loses money: its worst case is +\$1M, against −\$9M for the Large Hub.

CloudFlow capacity alternatives, lab Part B (B7:D11) | Net payoffs | Unit: millions of USD

How to read it: each row is a choice; each column is a demand state with its probability. EMV is the probability-weighted average of the row.



*Payoff Tables and
Expected Monetary Value*

WORKED EVIDENCE: REGIONAL HUB EXPANSION ANALYSIS

Payoff Tables and Expected Monetary Value

1

Net payoffs by demand state

- Large Hub: \$12M high, \$3M medium, -\$9M low. Modular Hub: \$9M, \$5M, \$1M.

2

Expected value results

- Large Hub EMV = $0.30(12) + 0.50(3) + 0.20(-9) = \3.30M . Modular Hub EMV = $0.30(9) + 0.50(5) + 0.20(1) = \5.40M .

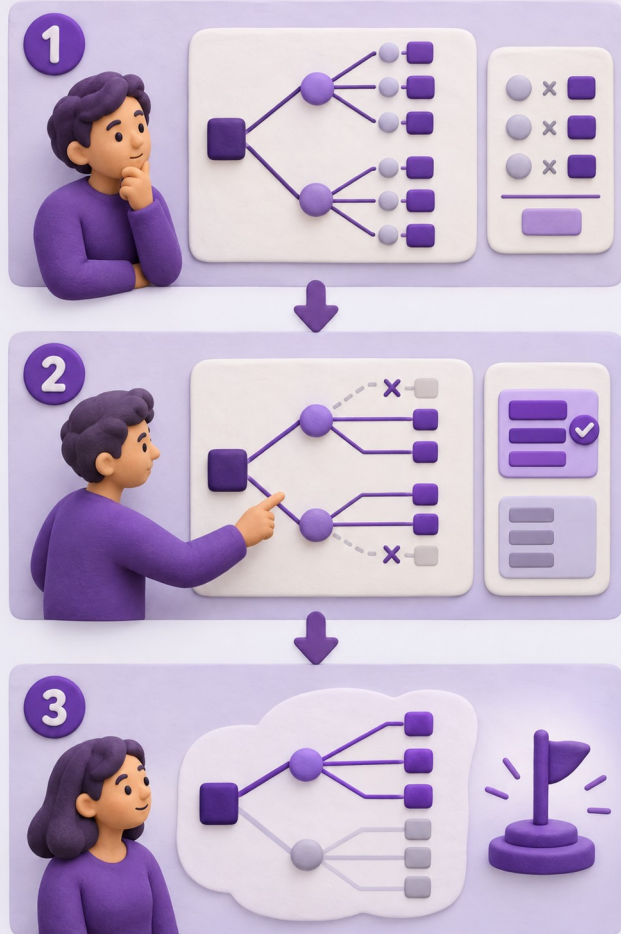
3

What it means

- The Modular Hub wins on EMV by \$2.10M and never loses money.
- The Large Hub would need a much higher chance of high demand to catch up.

THE ROLLBACK MECHANISM AND BACKWARD INDUCTION

Decision Trees and Risk Attitude



1

How it works

- **Step-by-Step Procedure - At Chance Nodes:** Compute **EMV** as the probability-weighted sum of future branch payoffs.
- **At Decision Nodes** Prune dominated alternatives and select the maximum expected value (or minimum expected cost). - At Root Node: Identify the optimal upfront strategic path under risk neutrality.

EXPECTED MONETARY VALUE CALCULATION (1/2)

$$EMV = \sum_{i=1}^n P(S_i) \times X_i$$

Symbols:

EMV: Expected Monetary Value of an uncertain decision branch (measured in currency units).

P(S_i): Discrete probability of state of nature i, where 0 ≤ P(S_i) ≤ 1 and sum of P(S_i) = 1.0.

X_i: Net economic payoff realized if state of nature i occurs.

Inputs: Probabilities P(S₁) = 0.30, P(S₂) = 0.50, P(S₃) = 0.20; Net payoffs X₁ = 12.0M, X₂ = 3.0M, X₃ = -\$9.0M.

Steps:

1. Multiply each probability by its net payoff: 0.30 * 12.0 = 3.60, 0.50 * 3.0 = 1.50, 0.20 * (-9.0) = -1.80.

EXPECTED MONETARY VALUE CALCULATION (2/2)

$$EMV = \sum_{i=1}^n P(S_i) \times X_i$$

Steps:

2. Sum the weighted products: $3.60 + 1.50 + (-1.80) = 3.30$ million dollars.

Result: EMV = \$3.30 million

The Large Hub yields an expected monetary value of \$3.30 million across repeated demand cycles, but carries substantial downside risk.

Use: Used at every chance node during backward induction to fold back multi-stage decision trees.

Limit: Assumes risk neutrality; does not account for variance or bankruptcy risk from extreme negative states.

EXCEL ROLLBACK CALCULATION WITH SUMPRODUCT (1/2)

```
=SUMPRODUCT(B2:D2, B3:D3)
```

Symbols:

B2:D2: Vector of discrete probabilities for states of nature (summing to 1.0).

B3:D3: Vector of net economic payoffs corresponding to each state of nature.

Inputs: Cells B2:D2 contain {0.30, 0.50, 0.20}; Cells B3:D3 contain {12.0, 3.0, -9.0}.

Steps:

1. Enter probabilities in range B2:D2.
2. Enter net payoffs in range B3:D3.
3. In cell E3, enter formula =SUMPRODUCT(B2:D2, B3:D3).

EXCEL ROLLBACK CALCULATION WITH SUMPRODUCT (2/2)

```
=SUMPRODUCT(B2:D2, B3:D3)
```

Steps:

4. Formula computes $(0.30 \times 12.0) + (0.50 \times 3.0) + (0.20 \times -9.0) = 3.30$.

Result: 3.30

Evaluates expected monetary value across states of nature in a single robust spreadsheet operation.

Use: Standard implementation for chance node evaluation in Excel decision tree templates.

Limit: Ranges must be identical in dimension and orientation; probability cells must sum exactly to 1.0.

WORKED EXAMPLE: CLOUDFLOW CAPACITY DECISION ROLLBACK

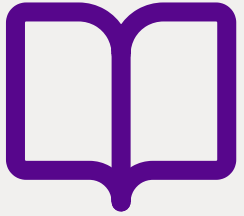
Decision Trees and Risk Attitude

Alternative	High demand (30%)	Medium demand (50%)	Low demand (20%)	EMV
Large Hub	12.00	3.00	−9.00	3.30
Modular Hub	9.00	5.00	1.00	5.40

The Modular Hub dominates with an expected monetary value of \$5.40 million versus \$3.30 million for the Large Hub, while avoiding catastrophic downside losses.

Regional freight hub capital allocation scenarios | 5-year planning horizon | Unit: Millions of USD

Rollback: at the decision node, compare the two EMVs and keep the larger (\$5.40M); prune the Large Hub branch. Then ask whether its worst case is acceptable.



*Decision Trees and Risk
Attitude*

WORKED EXAMPLE: ROLLING BACK THE HUB DECISION

Decision Trees and Risk Attitude

1

Payoffs and EMV

- Large Hub: \$12M high, \$3M medium, -\$9M low. $EMV = 0.30(12) + 0.50(3) + 0.20(-9) = \$3.30M$.
- Modular Hub: \$9M, \$5M, \$1M. $EMV = 0.30(9) + 0.50(5) + 0.20(1) = \$5.40M$.

2

Rollback decision

- Prune the Large Hub branch: its EMV is \$2.10M lower.
- Choose the Modular Hub: higher EMV, and it never loses money. EMV is a long-run average, not a guaranteed payoff.

AI SEARCH & SHARE



*Discuss with a partner;
Artificial intelligence (AI)
support is optional.*

APPLY THE IDEA

Discuss with a partner; Artificial intelligence (AI) support is optional.

1

Partner question

- When could a risk-averse committee rationally choose the option with the lower EMV?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

2

How to check an AI answer

- Ask it which assumption about risk its answer depends on.
- Test it on CloudFlow: the Modular Hub has the higher EMV and never loses money. Is there a trade-off here at all?

DECISION POINT: WHICH HUB SHOULD CLOUDFLOW BUILD?

Use the evidence, explain your decision, and name one limit.



Decision question

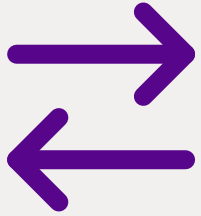
The Large Hub pays the most when demand is high. Why does the Modular Hub still have the higher EMV, and which would you recommend?

The evidence

Large Hub: \$12M, \$3M or -\$9M (EMV \$3.30M). Modular Hub: \$9M, \$5M or \$1M (EMV \$5.40M). High demand has a 30% chance; medium and low together, 70%.

Name one limit

EMV is a long-run average. In a single low-demand year the Large Hub loses \$9M.



Setup, then practice and explain

CLASS PRACTICE 2: COMPARE CAPACITY ALTERNATIVES WITH EMV

10 minutes in the lab, then the break (slide 38)

1

Before the practice timer

- **Instructor demo (3 min):** follow along in SPSS block H2 and leave the output open.
- **Then:** open **IN-CLASS_Lab Part B** and start the 10-minute timer. Focus on **D14:D17**.

2

Check and bring back

- **Check:** **D15** shows 5.40 (million dollars) and **D16** equals the larger EMV. **Bring back:** why an EMV is not a guaranteed payoff.

3

If you need help, or finish early

- **If you need help:** **D14** is =SUMPRODUCT(\$B\$7:\$D\$7,B10:D10). First check that **B7:D7** add up to 100%.
- **If you finish early:** on paper, set demand to 30% high, 20% medium, 50% low. Which hub has the higher EMV now?



Pause. Reset. Return ready.



BREAK

Save your workbook before leaving.

Stretch, Hydrate, and Reset

SECTION 3

MONTE CARLO SIMULATION AND THE FLAW OF AVERAGES

Show the flaw of averages in capacity models and build multi-trial Monte Carlo simulations in Excel.
In this section: the flaw of averages · building Excel simulations



*The Flaw of Averages in
Spreadsheet Models*

THE MECHANISM: HOW NON-LINEAR PAYOFFS BREAK SYMMETRY

The Flaw of Averages in Spreadsheet Models

1

Payoffs have limits

- Profit formulas use limits such as **MIN**(Capacity, Demand) and **MAX**(0, Capacity – Demand). Those limits bend the profit line, so it is no longer straight (it is concave).

2

The upside is capped

- Demand above capacity earns nothing extra, because every unit is already sold. Demand below capacity leaves unused units that salvage for less than they cost.

3

So the average plan misleads

- Here gains above average demand are cut off while losses below it are kept in full, so mean profit falls below profit at mean demand. (A convex model errs the other way.)

MATHEMATICAL FORMULATION OF THE FLAW OF AVERAGES (1/3)

$$\mathbb{E}[f(X)] \neq f(\mathbb{E}[X]) \quad \text{for non-linear } f(x)$$

Symbols:

X: Random variable representing uncertain business input (e.g., customer demand, clearance days)

f(x): Payoff or cost function mapping input realizations to financial outcomes

E[X]: Expected value or population mean of the input random variable

E[f(X)]: True expected monetary outcome averaged across all states of nature

f(E[X]): Naive deterministic outcome calculated by evaluating the function at the average input

MATHEMATICAL FORMULATION OF THE FLAW OF AVERAGES (2/3)

$$\mathbb{E}[f(X)] \neq f(\mathbb{E}[X]) \quad \text{for non-linear } f(x)$$

Inputs: Input distribution of X with probability mass function $P(X = x_i)$ and non-linear payoff formula $f(x)$.

Steps:

1. Compute expected input $\mathbb{E}[X] = \sum_{i=1}^n x_i P(X = x_i)$.
2. Evaluate naive outcome by plugging $\mathbb{E}[X]$ directly into $f(x)$ to obtain $f(\mathbb{E}[X])$.
3. Evaluate state-by-state outcomes $f(x_i)$ for each realized state i .
4. Compute true expected outcome $\mathbb{E}[f(X)] = \sum_{i=1}^n f(x_i) P(X = x_i)$.
5. Compute valuation gap $\Delta = |f(\mathbb{E}[X]) - \mathbb{E}[f(X)]|$.

MATHEMATICAL FORMULATION OF THE FLAW OF AVERAGES (3/3)

$$\mathbb{E}[f(X)] \neq f(\mathbb{E}[X]) \quad \text{for non-linear } f(x)$$

Result: True expected outcome $\mathbb{E}[f(X)]$ differs from deterministic evaluation $f(\mathbb{E}[X])$.

Whenever $f(x)$ is strictly concave (such as capacity-capped revenue), $\mathbb{E}[f(X)] < f(\mathbb{E}[X])$. Whenever $f(x)$ is strictly convex (such as penalty costs with free tiers), $\mathbb{E}[f(X)] > f(\mathbb{E}[X])$.

Use: Inventory management, service queue sizing, demurrage budgeting, and financial derivatives pricing.

Limit: The inequality does not quantify the magnitude of the distortion without specifying the variance of X and curvature of $f(x)$.

EXCEL FORMULAS FOR BOUNDED INVENTORY PAYOFFS (1/3)

```
=MIN(Order_Qty, Demand_Actual) * Unit_Price + MAX(0, Order_Qty - Demand_Actual) *  
Unit_Salvage - (Order_Qty * Unit_Cost)
```

Symbols:

Order_Qty: Cell reference to planned purchase order volume (e.g., \$B\$1)

Demand_Actual: Cell reference to realized customer demand (e.g., C4)

Unit_Price: Retail selling price per unit (e.g., \$B\$2)

Unit_Salvage: Liquidation recovery value per unsold unit (e.g., \$B\$3)

Unit_Cost: Procurement cost per unit (e.g., \$B\$4)

Inputs: Order_Qty = 1000, Unit_Price = 100, Unit_Salvage = 15, Unit_Cost = 40, Demand_Actual in {600, 1000, 1400}.

Steps:

EXCEL FORMULAS FOR BOUNDED INVENTORY PAYOFFS (2/3)

```
=MIN(Order_Qty, Demand_Actual) * Unit_Price + MAX(0, Order_Qty - Demand_Actual) *  
Unit_Salvage - (Order_Qty * Unit_Cost)
```

Steps:

1. Enter formula in cell D4: $\text{=MIN}(\$B\$1, C4) * \$B\$2 + \text{MAX}(0, \$B\$1 - C4) * \$B\$3 - (\$B\$1 * \$B\$4)$
2. For $C4 = 600$: $\text{MIN}(1000, 600) * 100 + \text{MAX}(0, 400) * 15 - (1000 * 40) = 60000 + 6000 - 40000 = 26000$
3. For $C5 = 1000$: $\text{MIN}(1000, 1000) * 100 + \text{MAX}(0, 0) * 15 - (1000 * 40) = 100000 + 0 - 40000 = 60000$
4. For $C6 = 1400$: $\text{MIN}(1000, 1400) * 100 + \text{MAX}(0, -400) * 15 - (1000 * 40) = 100000 + 0 - 40000 = 60000$

EXCEL FORMULAS FOR BOUNDED INVENTORY PAYOFFS (3/3)

```
=MIN(Order_Qty, Demand_Actual) * Unit_Price + MAX(0, Order_Qty - Demand_Actual) *  
Unit_Salvage - (Order_Qty * Unit_Cost)
```

Steps:

5. Compute expected profit in cell D7: =SUMPRODUCT(D4:D6, E4:E6) where E4:E6 contains probabilities {0.3333, 0.3333, 0.3333}.

Result: Expected Monetary Value = 48666.67 USD.

Calculates profit under demand uncertainty, explicitly incorporating capacity ceilings and liquidation discounts.

Use: Retail replenishment models, seasonal merchandise procurement, and warehouse allocation.

Limit: Assumes constant unit prices and salvage values that do not decline further as liquidated volume rises.

WORKED EVIDENCE: CLOUDFLOW PROFIT UNDER DEMAND UNCERTAINTY

The Flaw of Averages in Spreadsheet Models

Demand State	Demand (Units)	Actual Sales	Salvage Units	Net Profit (USD)	Probability
Low	600	600	400	26000	0.3333
Medium	1000	1000	0	60000	0.3333
High	1400	1000	0	60000	0.3333
Expected Value	1000	866.67	133.33	48666.67	1

Deterministic modeling at expected demand overstates true expected profit by \$11,333.33 because stockouts truncate upside while unsold units incur steep losses.

Winter inventory planning states | Q4 seasonal inventory cycle | Unit: USD



*The Flaw of Averages in
Spreadsheet Models*

WORKED EVIDENCE: CLOUDFLOW CAPACITY UNDER UNCERTAIN DEMAND

The Flaw of Averages in Spreadsheet Models

1

Worked example: three demand levels

- CloudFlow commits to 1,000 capacity units at \$40 each, sells at \$100 and salvages unused units at \$15.
- Demand is equally likely to be 600, 1,000 or 1,400 units, so expected demand is exactly 1,000 units.

2

Average plan versus expected value

- At the mean demand of 1,000 units, profit is \$60,000.
- Averaged over the three equally likely levels, expected profit (EMV) is \$48,666.67.

3

The gap

- Planning at the average overstates expected profit by \$11,333.33 and hides the loss when demand is low.



Building Excel Monte Carlo Simulations

HOW THE LAB DRAWS 1,000 TRIALS

Building Excel Monte Carlo Simulations

1

How it works: Inverse Transform Sampling

- **Data_Trials** holds 1,000 fixed probabilities between 0 and 1. Everyone uses the same ones, so the Check-your-work boxes match exactly.

2

Demand and profit in row 32

- **Demand: B32** turns the first fixed probability into demand:
=NORM.INV(Data_Trials!B2,\$B\$7,\$B\$8)
- **Profit: C32** applies the B14 logic to that demand:
=MIN(B32,\$B\$9)*\$B\$10+MAX(\$B\$9-B32,0)*\$B\$11-\$B\$9*\$B\$12

3

1,000 trials, then the summaries

- Fill **B32:C32** down to row 1031. **B16:B18** give the mean, the SD (**STDEV.S**) and the 5th percentile (Excel **PERCENTILE.INC**).
- **Optional live route** (**LAB02_Live_Exploration.xlsx**, **RAND()**): numbers change on every recalculation, so they will not match the checks.

NORMAL INVERSE TRANSFORMATION FOR CONTINUOUS STOCHASTIC VARIATES (1/3)

$$X = F^{-1}(U) = \mu + \sigma \cdot \Phi^{-1}(U)$$

Symbols:

X: The realized random variate generated for the simulation trial.

U: A probability between 0 and 1. In the lab it is a fixed Data_Trials value; RAND() appears only in the optional live workbook, whose numbers change on every recalculation.

F⁻¹: The inverse cumulative distribution function of the target continuous distribution.

μ: The expected value or mean of the underlying population distribution.

σ: The standard deviation measuring dispersion around the mean.

Φ⁻¹: The standard normal inverse cumulative distribution function, implemented in Excel via NORM.S.INV().

NORMAL INVERSE TRANSFORMATION FOR CONTINUOUS STOCHASTIC VARIATES (2/3)

$$X = F^{-1}(U) = \mu + \sigma \cdot \Phi^{-1}(U)$$

Inputs: Mean customer demand = 1,000 units, standard deviation = 250 units, and probability $U = 0.3520$.

Steps:

1. Evaluate the standard normal inverse cumulative function at U : $\text{NORM.S.INV}(0.3520) = -0.3799$.
2. Scale the standard z-score by the target standard deviation: $-0.3799 \cdot 250 = -94.975$ units.
3. Shift the scaled dispersion by the target distribution mean: $1,000 + (-94.975) = 905.025$ units.
4. Combine the operations directly in Excel syntax: $=\text{NORM.INV}(0.3520, 1000, 250) = 905.02$ units.

NORMAL INVERSE TRANSFORMATION FOR CONTINUOUS STOCHASTIC VARIATES (3/3)

$$X = F^{-1}(U) = \mu + \sigma \cdot \Phi^{-1}(U)$$

Result: 905.025 units

A probability of 0.3520 translates to a simulated seasonal demand of approximately 905 units, sitting 0.38 standard deviations below the expected mean.

Use: Lab cell B32 applies it to each fixed probability: `=NORM.INV(Data_Trials!B2,$B$7,$B$8)`. Optional live route (LAB02_Live_Exploration.xlsx): numbers change on every recalculation, so they will not match the checks.

Limit: Assumes the underlying random variable is symmetrically distributed and unbounded, which can produce theoretically negative quantities if the mean is within three standard deviations of zero.

THE LAB'S TRIAL ROWS: DEMAND AND PROFIT (1/3)

B32: =NORM.INV(Data_Trials!B2,\$B\$7,\$B\$8)

C32: =MIN(B32,\$B\$9)*\$B\$10+MAX(\$B\$9-B32,0)*\$B\$11-\$B\$9*\$B\$12

Fill B32:C32 down to row 1031 → 1,000 trials

Symbols:

Data_Trials!B2: the first of 1,000 fixed probabilities between 0 and 1; row 33 reads B3, and so on.

\$B\$7, \$B\$8: mean demand (1,000) and SD (250), locked with \$ so they stay put when you fill down.

\$B\$9 to \$B\$12: capacity (1,000 units), price (\$100), salvage (\$15) and acquisition cost (\$40).

MIN(B32,\$B\$9): units sold, capped at capacity. **MAX(\$B\$9-B32,0):** unused units, sold for salvage.

\$B\$9*\$B\$12: the cost of every unit stocked, paid whatever the demand.

THE LAB'S TRIAL ROWS: DEMAND AND PROFIT (2/3)

B32: `=NORM.INV(Data_Trials!B2,$B$7,$B$8)`

C32: `=MIN(B32,$B$9)*$B$10+MAX($B$9-B32,0)*$B$11-$B$9*$B$12`

Fill B32:C32 down to row 1031 → 1,000 trials

Inputs: Data_Trials!B2 = 0.08307 (trial 1); B7 = 1,000; B8 = 250; B9 = 1,000; B10 = \$100; B11 = \$15; B12 = \$40.

Steps:

1. B32 = $\text{NORM.INV}(0.08307, 1000, 250) = 653.82$ units of demand.
2. Units sold = $\text{MIN}(653.82, 1000) = 653.82$; unused = $\text{MAX}(1000 - 653.82, 0) = 346.18$.
3. C32 = $653.82 \times 100 + 346.18 \times 15 - 1000 \times 40 = \$30,574.76$.

Result: trial 1 earns \$30,574.76. If row 32 shows these values, your formulas are right.

THE LAB'S TRIAL ROWS: DEMAND AND PROFIT (3/3)

B32: `=NORM.INV(Data_Trials!B2,$B$7,$B$8)`

C32: `=MIN(B32,$B$9)*$B$10+MAX($B$9-B32,0)*$B$11-$B$9*$B$12`

Fill B32:C32 down to row 1031 → 1,000 trials

Fill B32:C32 down to row 1031. Each row reads the next Data_Trials probability, so rows 32 to 1031 hold 1,000 trials.

Result: B20 counts 1,000 numeric trials; B16:B19 summarize them (next slide).

Use: every student has the same 1,000 trials, so the Check-your-work boxes and the checkpoint in B1033:H1035 work.

Limit: 1,000 trials are one sample. Their mean (\$51,319.75) differs from the exact expected profit (\$51,522.48) by sampling error.

Optional live route (LAB02_Live_Exploration.xlsx): demand `=NORM.INV(RAND(),$B$7,$B$8)` with a What-If Data Table. Numbers change on every recalculation, so they will not match the checks.

WORKED SIMULATION: CLOUDFLOW CAPACITY PROFIT

Lab Part C assumptions: demand mean 1,000, SD 250; capacity 1,000; price \$100; salvage \$15; cost \$40

Metric	Plan at mean demand	1,000 simulated trials	Difference
Expected profit	\$60,000	\$51,320	-\$8,680
Standard deviation	\$0	\$12,701	+\$12,701
5th-percentile profit	\$60,000	\$25,884	-\$34,116
Maximum profit	\$60,000	\$60,000	\$0
Loss probability	0.0%	0.4%	+0.4 points

Planning at average demand overstates expected profit by \$8,680: sales are capped when demand is high, while unused capacity loses money when demand is low.

CloudFlow capacity model (lab Part C, B7:B12) | 1,000 fixed reference trials (Data_Trials) | Unit: USD



*Building Excel Monte
Carlo Simulations*

WHAT THE SIMULATION SHOWS

Building Excel Monte Carlo Simulations

1

The CloudFlow capacity model

- CloudFlow commits to 1,000 capacity units at \$40 each, sells at \$100 and salvages unused units at \$15.
- Demand is normal with mean 1,000 and SD 250 units; planning at the mean gives \$60,000 profit.

2

Results across 1,000 trials

- Mean profit \$51,320; standard deviation \$12,701; 5th-percentile profit \$25,884.

3

The flaw of averages

- Capacity caps profit at \$60,000, while each unused unit loses \$25 (\$40 cost – \$15 salvage).
- Expected profit falls \$8,680 below the plan, and 0.4% of trials lose money.

AI SEARCH & SHARE



*Discuss with a partner;
Artificial intelligence (AI)
support is optional.*

APPLY THE IDEA

Discuss with a partner; Artificial intelligence (AI) support is optional.

1

Partner question

- How should an executive interpret a simulation output showing an expected profit of \$51,320 with a 5th-percentile profit of \$25,884?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

2

How to check an AI answer

- Ask it to say what the 5th percentile means in one plain sentence.
- Test it on CloudFlow: in 5% of trials profit falls below \$25,884. What does that mean for one planning year?

DECISION POINT: WHY DOES THE AVERAGE PLAN MISLEAD?

Use the evidence, explain your decision, and name one limit.



Decision question

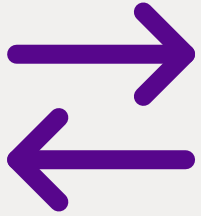
Why does profit at average demand (\$60,000) overstate the average simulated profit, and what should CloudFlow plan around?

The evidence

Profit at mean demand: \$60,000. Mean of 1,000 trials: \$51,320 (SD \$12,701). 5th-percentile profit: \$25,884. Gap: \$8,680.

Name one limit

The model assumes normal demand (mean 1,000, SD 250), capacity of 1,000 units at \$40 and \$15 salvage.



Setup, then practice and explain

CLASS PRACTICE 3: BUILD THE REPRODUCIBLE PROFIT SIMULATION

12 minutes in the lab, then back to slide 61: the group studio

1

Before the practice timer

- **Instructor demo (5 min):** follow along in SPSS block H3 and leave the output open.
- **Then:** open **IN-CLASS_Lab Part C**. Use the fixed **Data_Trials** inputs and fill **B32:C32** down to row 1031.

2

Check and bring back

- **Check:** **B20** = 1,000 trials, **B16:B18** are numbers, and no cell shows an error. Timer 12 min, debrief 2 min.

3

If you need help, or finish early

- **If you need help:** compare your **C32** with the profit formula in **B14**; only the demand cell should change.
- **If you finish early:** predict, without editing the model, how **B18** would move if the demand SD rose from 250 to 350. Why?

SECTION 4

GROUP STUDIO: ADVISE CLOUDFLOW'S COMMITTEE

Bring Sections 1 to 3 together in your group and give the committee a recommendation.
In this section: the brief · your evidence · four lenses · the studio · share and vote

THE STUDIO BRIEF

PNP-GRP page 1, in short. Your group gives the committee its final answer.



Situation

CloudFlow must decide whether to fund modular support capacity. Leaders disagree: do ticket flags predict who will cancel, or waste engineers on accounts that would stay?

A

Commit and reach out

Reserve capacity now; call every high-ticket account. Benefit: fast fixes. Risk: wasted calls to loyal accounts.

B

Hold and review

Keep capacity steady; managers review flags first. Benefit: capital stays flexible. Risk: slower help loses accounts.

Your group's job

Scene: *The committee meets Monday at 9. The chair wants your group's advice on one page, ready to act on.*

In the PNP-GRP, answer Q1 and your lens's Q2. Each question has its own two options: pick one, give a reason and one unintended consequence, 30 to 60 words each. No new calculations.

YOUR EVIDENCE: WHAT YOU CHECKED TODAY

No new calculations in the studio: cite these results with their units and denominators

Lab part	What you found	What it cannot tell you
Part A · Section 1 (B18)	Of 272 flagged accounts, 96 churned: 35.3%, against a 12.0% baseline. 176 flagged accounts renewed.	Whether outreach would stop a flagged account from leaving.
Part B · Section 2 (D14:D17)	Modular Hub EMV \$5.40M against \$3.30M for the Large Hub; the Large Hub loses \$9M when demand is low.	What happens in one bad year: EMV is a long-run average.
Part C · Section 3 (B16:B20)	Plan at mean demand: \$60,000. Average of 1,000 trials: \$51,320; 5th percentile \$25,884.	Whether next year's demand will follow the same model.
SPSS H4 · downside check	4 of the 1,000 trials lose money (0.4%).	How large a loss the committee can absorb.

Evidence checker: before the Recorder types, confirm that each number your group cites is in this table or in your workbook.

Source: your IN-CLASS_Lab Parts A to C and SPSS blocks H1 to H4 | Units: accounts, probabilities, \$ millions (EMV) and dollars (profit)

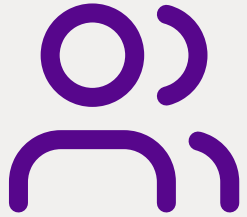
FOUR LENSES, ONE DECISION

Everyone answers Q1; each group's Q2 comes from its lens (PNP-GRP pages 3, 5, 7 and 9)

Group	Lens	Your Q2
1	Customer experience	When does ticket-based screening fail to protect client relationships?
2	Financial controller	When is committing upfront modular support capital financially justified?
3	Capacity planning	Where does planning with average ticket demand, rather than simulation trials, mislead staffing?
4	Data quality	When does data from past accounts fail to predict future churn?

Q1 for every group: which misjudgment is more dangerous for CloudFlow, wasting support capital on accounts that would stay, or losing frustrated accounts that needed help?

Each Q2 draws on one section: Section 1 (groups 1 and 4), Section 2 (group 2), Section 3 (group 3)



*Think alone, decide
together, share*

RUN THE STUDIO: 28 MINUTES

Open the PNP-GRP now and find your group's pages

1

Before the groups (5 min)

- **Instructor demo (2 min):** SPSS block H4, the downside check.
- **Think alone (3 min):** in **B1033** of Lab Part C, write one principle and one condition that would reverse it.

2

In your group (13 min)

- **Roles (3 min):** Facilitator, Evidence checker, Recorder, Reporter.
- **Answer (10 min):** Q1 and your lens's Q2. Each has its own two options: pick one, give a reason and one unintended consequence.

3

Share and vote (10 min)

- **Share (7 min):** each Reporter gives one insight in about 45 seconds.
- **Vote (3 min):** the class advises the committee on the next slide.

SHARE AND VOTE: ADVISE THE COMMITTEE

Hear every lens, then vote. A conditional answer is allowed.

Scene: *Monday, 9 a.m., in the boardroom. The chair has promised to follow the class vote, so every hand counts.*

Which option should the committee take?

- **A: Commit and reach out**

Reserve capacity now; call every high-ticket account. Best if lost accounts cost far more than calls.

- **B: Hold and review**

Keep capacity steady; review flags first. Best if most flags are false alarms (176 of 272).

- **C: Commit only after a pilot**

Pilot outreach first; commit if helped accounts stay. Best if outreach's effect is still uncertain.



Answer the committee

WHAT TODAY'S EVIDENCE SUPPORTS

Four tools, one conditional recommendation

1

What each tool told us

- The contingency table set the right denominator. Bayes rule lifted churn risk from 12% to 35.3% after a flag. EMV favored the Modular Hub. Simulation showed that the average plan overstates profit.
- **The limit:** high tickets go with churn; that does not prove CloudFlow's outreach will keep accounts.

2

A defensible recommendation

- Commit only if a pilot shows that outreach keeps flagged accounts and the downside fits the committee's risk tolerance; otherwise hold and review.



Save, resume, reflect

CONTINUE FROM YOUR SAVED CHECKPOINT

The Take home portion completes this session's lab

1

Save before leaving

- Save your workbook with your name and NetID before you leave.

2

Resume at home

- Take Home 1: Multi-Year Capital Investment Risk and Sensitivity.
- Resume at **POST-CLASS_Part 1!A7**. Preview the model in Appendix D (slides 90 to 98).

3

Check and submit

- Complete the listed checks and submit the workbook and memo. The LAB02 guide gives every formula; only the written choices are yours.



*One optional question
remains open.*

THANK YOU AND LOOKING AHEAD

One optional question remains open.

1

Thank you

- Thank you for your analysis today.

2

Optional reflection, not homework

- What additional CloudFlow evidence would most reduce uncertainty before leaders authorize the screening intervention and modular capacity commitment?



*Everything used today,
and where to find it*

SESSION REFERENCES

Everything used today, and where to find it

1

In class today

- **Slides and notes:** Session 04, the CloudFlow Analytics case (this deck).
- **Lab:** LAB02 guide, **LAB02_Learning_Workbook.xlsx**, **LAB02_SPSS.sps** and the data file **cloudflow_churn_cohort.csv**.
- **Optional live route** (**LAB02_Live_Exploration.xlsx**): numbers change on every recalculation, so they will not match the checks.

2

After class

- **Concise Toolkit:** Section 1 pp. 3-8, Section 2 pp. 9-16, Section 3 pp. 17-23, Section 4 pp. 24-29.
- **Take-home:** **POST-CLASS_Part 1** in your LAB02 workbook, with the memo and the AI-use disclosure.

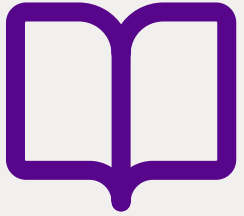
APPENDIX

APPENDIX A: SELECTED STUDY SUMMARIES

Selected examples for further study. Find complete explanations, worked examples and references under the matching topic in the Session Study Toolkit.

Key point

Selected examples for further study. Find complete explanations, worked examples and references under the matching topic in the Session Study Toolkit.



*Payoff Tables and
Expected Monetary Value*

COMMON MISCONCEPTION: TREATING EMV AS A GUARANTEED SINGLE OUTCOME

Payoff Tables and Expected Monetary Value

1

The Average Fallacy

- **EMV** is a long-run mathematical expectation across repeated trials, not a guaranteed return on an individual project.

2

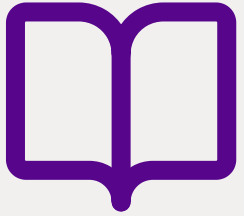
Realized Outcomes Differ

- For CloudFlow, the Modular Hub yields \$9.0M, \$5.0M, or \$1.0M. The calculated \$5.40M average is physically impossible in any single execution.

3

Reporting Discipline

- Always present the full range of potential payoffs alongside **EMV** to avoid masking downside solvency risks from executive leadership.



*Decision Trees and Risk
Attitude*

COMMON MISCONCEPTIONS IN TREE CONSTRUCTION AND VALUATION

Decision Trees and Risk Attitude

1

Common Decision Modeling Errors

- **Chronology Inversion:** Placing decision nodes after chance nodes when capital must be committed prior to resolution.

2

Guaranteed Value Fallacy:

- Believing **EMV** represents the single-project outcome rather than an expected average.

3

Gross vs Net Accounting:

- Omitting upfront sunk investment costs from terminal branch payoffs.
- **Probability Misallocation:** Failing to ensure chance branches are mutually exclusive and sum to 1.0.



*Probability Essentials and
Contingency Tables*

THE DOUBLE-COUNTING ADDITION FALLACY

Probability Essentials and Contingency Tables

1

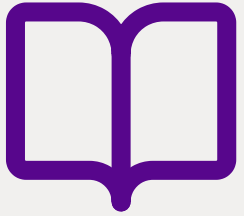
Common Error: Naive Addition

- Summing $P(\text{Churn}) + P(\text{HighTickets}) = 0.120 + 0.272 = 0.392$ double-counts overlapping clients, distorting exposure upward.

2

Correct Correction

- Because events co-occur, subtract the intersection: $0.392 - 0.096 = 0.296$. Pure addition applies only when events are mutually exclusive.



*Conditional Probability
and Bayesian Updating*

THE TRANSPOSITION FALLACY AND COMMON MISCONCEPTIONS

Conditional Probability and Bayesian Updating

1

The Transposition Fallacy

- Confusing $P(A|B)$ with $P(B|A)$ is the most dangerous error in diagnostic business screening.

2

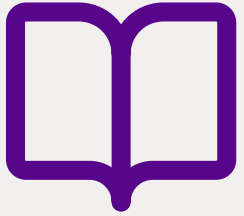
Clear Distinction

- $P(\text{Signal}|\text{Risk})$ evaluates only known failures; $P(\text{Risk}|\text{Signal})$ evaluates all accounts generating that alert.

3

Why It Matters

- Assuming a 90% overdraft rate among defaulters means 90% of overdraft accounts will default leads to massive over-rejection of viable applicants.



*Lab 2 Guided
Implementation and
Sensitivity Analysis*

COMMON TECHNICAL ERRORS IN SPREADSHEET SIMULATION

Lab 2 Guided Implementation and Sensitivity Analysis

1

Data Table freezing (optional route only)

- **Optional live route** (**LAB02_Live_Exploration.xlsx**): numbers change on every recalculation, so they will not match the checks.
- There, a Data Table whose column input points at a model cell, not a blank cell, repeats identical values. The lab's fixed **Data_Trials** rows need no Data Table.

2

Simulation Trials vs. Empirical Samples

- A simulation trial count (e.g., 1,000 iterations) measures computational precision of the assumed distribution.
- It does not represent empirical field data or validate real-world model accuracy.



*Common Misconception:
The Flaw of Averages in
Spreadsheet Models*

THE LAW OF LARGE NUMBERS DOES NOT CURE NON-LINEARITY

Common Misconception: The Flaw of Averages in Spreadsheet Models

1

Common Misconception

- Believing that multi-store aggregation or long-run repetition eliminates the Flaw of Averages by averaging out fluctuations.

2

Theoretical Reality

- The Law of Large Numbers guarantees convergence of profits to true expected value (\$48,666.67), not the naive deterministic model (\$60,000).

3

Practical Warning

- Operational constraints operate locally at each store. Truncated sales and salvage write-downs accumulate across chains, locking in the shortfall.

APPENDIX

APPENDIX B: EXPLORATION AND STRETCH

Optional independent study

Key point

Optional selected readings: Exponential Utility; Utility Example

EXPONENTIAL UTILITY (1/2)

$$U(x) = 1 - e^{-x/R}$$

$$E[U(X)] = \sum_i p_i U(x_i)$$

$$CE = -R \ln(1 - E[U(X)])$$

$$RP = EMV - CE$$

Symbols:

x is net monetary payoff.

R is risk tolerance in the same monetary units.

CE is certainty equivalent and RP is risk premium.

Inputs: Outcome probabilities, monetary payoffs, and risk tolerance R.

Steps:

1. Transform each payoff into utility.
2. Weight utilities by probabilities and sum.

EXPONENTIAL UTILITY (2/2)

$$U(x) = 1 - e^{-x/R}$$

$$E[U(X)] = \sum_i p_i U(x_i)$$

$$CE = -R \ln(1 - E[U(X)])$$

$$RP = EMV - CE$$

Steps:

3. Invert expected utility to obtain CE, then subtract CE from EMV.

Result: CE expresses the risky prospect as an equivalent sure amount.

A positive RP quantifies the value assigned to avoiding risk.

Use: Risk-adjusted capital and decision-tree evaluation.

Limit: Constant absolute risk tolerance may not fit decisions spanning very different capital scales.

UTILITY EXAMPLE

A hypothetical venture valued by a risk-averse board

Outcome	Probability	Payoff \$M	Utility	Weighted utility
Gain	0.6	10	0.4866	0.292
Loss	0.4	-5	-0.3956	-0.1582
Total	1	EMV 4.00	E[U]	0.1338

The probability-weighted utilities produce a certainty equivalent below expected monetary value.

Two possible venture outcomes | Single investment decision | Unit: Probabilities, USD millions, and utility units



*A hypothetical venture
valued by a risk-averse
board*

UTILITY EXAMPLE

A hypothetical venture valued by a risk-averse board

1

Inputs

- Payoffs are +\$10 million with probability 0.60 and −\$5 million with probability 0.40. Risk tolerance is $R = 15$ million.

2

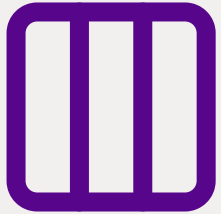
Expected value and utility

- $EMV = 0.60(10) + 0.40(-5) = \$4.00M$
- $E[U] \approx 0.1338$

3

Certainty equivalent and risk premium

- $CE = -15 \ln(1 - 0.1338) \approx \$2.15M$
- $RP = 4.00 - 2.15 \approx \$1.85M$



*A hypothetical venture
valued by a risk-averse
board*

UTILITY EXAMPLE | CONTINUED 2

A hypothetical venture valued by a risk-averse board

1

Interpretation

- Although the venture has a \$4.00 million expected value, this board values it like approximately \$2.15 million in certain cash.
- The \$1.85 million difference is the price assigned to avoiding volatility.

APPENDIX

APPENDIX C: INDUSTRY PRACTICE

Optional independent study

Key Point

Optional selected readings: GAO Cost-Risk Analysis; GAO Cost-Risk Analysis | Continued 2; Reading a P80 Budget; Chevron's DA Culture; Chevron's DA Culture | Continued 2



GAO COST-RISK ANALYSIS

From simulated costs to an S-curve

1

GAO setting

- The **U.S.** Government Accountability Office (**GAO**) publishes cost-estimating guidance for federal programs. Its March 2020 guide includes risk and uncertainty analysis in a credible estimate.

2

Measure and interpretation

- Analysts assign distributions to cost drivers, sample them repeatedly, and aggregate total program cost into a cumulative S-curve.
- **P80** is the 80th percentile of simulated total cost, in dollars: 80% of modeled outcomes are at or below it.



GAO COST-RISK ANALYSIS | CONTINUED 2

From simulated costs to an S-curve

1

Measure and interpretation

- The population is the simulations for one defined program scope, price basis, and reporting period - not all federal programs.

2

Limit and source

- Results inherit input, correlation, and scope assumptions. Source: GAO-20-195G, March 2020, <https://www.gao.gov/products/gao-20-195g>.



READING A P80 BUDGET

Illustrative contingency calculation

1

Illustrative FY2027 example

- For one hypothetical acquisition, base lifecycle cost is \$110 million in nominal dollars.
- Ten equally weighted simulated totals, sorted in \$ millions, are 96, 100, 103, 107, 111, 116, 121, 128, 138, 152.

2

Calculation and meaning

- Nearest-rank **P80** position = $\text{ceiling}(0.80 \times 10) = 8$; **P80** = \$128 million. Contingency above base = \$128M - \$110M = \$18M.
- Thus, 8 of 10 modeled outcomes fall at or below \$128M; this is not an 80% guarantee for actual cost.

3

Limit and source

- Ten trials are too few for practice, and unmodeled correlation can understate tails. Method source: GAO-20-195G, March 2020, <https://www.gao.gov/products/gao-20-195g>.



CHEVRON'S DA CULTURE

Organizational adoption of decision analysis

1

Reported adoption

- Chevron uses decision analysis (**DA**) to structure capital choices under uncertainty.
- A May 2018 **INFORMS** profile reported proprietary decision-analysis software used by more than 1,300 practitioners and described **DA** training for new hires.

2

Reported adoption

- The unit is practitioners using the system at the article's reporting time, not projects completed or dollars saved.



CHEVRON'S DA CULTURE | CONTINUED 2

Organizational adoption of decision analysis

1

Measure and use

- Adoption counts indicate organizational reach. Decision trees make choices, chance events, probabilities, and net payoffs explicit before expected monetary value is rolled back.

2

Evidence limit

- The article does not disclose active-project payoffs, drilling probabilities, or causal performance gains. Source: **INFORMS**, May 7, 2018, <https://pubsonline.informs.org/doi/10.1287/orms.2018.03.02/full/>.



*Lab 2 Guided
Implementation and
Sensitivity Analysis*

BUILDING THE SIMULATION ARCHITECTURE IN EXCEL

Appendix D: take-home preview, not presented in class

1

Layer 1: parameter block

- Centralizes capital outlay, fixed overhead, unit margin, discount rate, and demand parameters in dedicated labelled cells.
- **Trial Calculation:** Part C uses =**NORM.INV**(Data_Trials!**B2**, \$B\$7, \$B\$8); the capital model uses =**NORM.S.INV**(Data_Trials!**B2**).

2

Layer 2: reference trial sweep

- **Reference Trial Sweep:** Fixed Data_Trials inputs feed 1,000 rows, allowing exact checks without volatile recalculation or a blank workspace cell.

PRACTICE CASE: CASH FLOW AND SIMULATION FORMULAS (1/4)

```
=NORM.INV((A12-0.5)/1000, $B$4, $B$5)  
=NPV($B$7, C12:E12) - $B$2  
=COUNTIF(Sim_NPV, "<0") / COUNT(Sim_NPV)  
=PERCENTILE.INC(Sim_NPV, 0.05)
```

Symbols:

(A12-0.5)/1000: the fixed probability for trial i, with i = 1 to 1,000 in A12:A1011. The values are evenly spaced from 0.0005 to 0.9995, so anyone can reproduce the results exactly.

NORM.INV(prob, mean, stdev): the demand whose cumulative probability is prob.

NPV(rate, values): the present value of the Year 1 to 3 cash flows at a constant rate.

COUNTIF(range, criteria): counts the cells in a range that meet the condition.

PRACTICE CASE: CASH FLOW AND SIMULATION FORMULAS (2/4)

```
=NORM.INV((A12-0.5)/1000, $B$4, $B$5)  
=NPV($B$7, C12:E12) - $B$2  
=COUNTIF(Sim_NPV, "<0") / COUNT(Sim_NPV)  
=PERCENTILE.INC(Sim_NPV, 0.05)
```

Symbols:

PERCENTILE.INC(range, k): the k-th percentile of a range, with k between 0 and 1 inclusive.

Inputs: \$B\$2 capital outlay (\$2,500,000); \$B\$3 annual fixed overhead (\$600,000); \$B\$4 demand mean (4,500 seats); \$B\$5 demand SD (800); \$B\$6 contribution per seat (\$400); \$B\$7 discount rate (0.08). Practice case (not your take-home): each trial's demand is held for all three years.

Steps:

1. In A12:A1011 enter the trial numbers 1 to 1,000. In B12 enter =NORM.INV((A12-0.5)/1000, \$B\$4, \$B\$5) for that trial's demand.

PRACTICE CASE: CASH FLOW AND SIMULATION FORMULAS (3/4)

```
=NORM.INV((A12-0.5)/1000, $B$4, $B$5)  
=NPV($B$7, C12:E12) - $B$2  
=COUNTIF(Sim_NPV, "<0") / COUNT(Sim_NPV)  
=PERCENTILE.INC(Sim_NPV, 0.05)
```

Steps:

2. In C12:E12, compute each year's cash flow with $=\$B12 * \$B\$6 - \$B\$3$ ($\$B\6 = contribution per seat, $\$B\3 = annual fixed overhead).
3. In F12, compute the trial's NPV with $=NPV(\$B\$7, C12:E12) - \$B\2 .
4. Fill A12:F12 down to row 1011, so Sim_NPV (F12:F1011) holds 1,000 trial NPVs. No Data Table is needed.
5. Compute the loss probability in H4 with $=COUNTIF(Sim_NPV, "<0") / COUNT(Sim_NPV)$.

PRACTICE CASE: CASH FLOW AND SIMULATION FORMULAS (4/4)

```
=NORM.INV((A12-0.5)/1000, $B$4, $B$5)  
=NPV($B$7, C12:E12) - $B$2  
=COUNTIF(Sim_NPV, "<0") / COUNT(Sim_NPV)  
=PERCENTILE.INC(Sim_NPV, 0.05)
```

Steps:

6. Compute the 5th-percentile NPV in H5 with =PERCENTILE.INC(Sim_NPV, 0.05): about 95% of trials do better and about 5% do worse, so it is a cut-off, not a floor.

Result (Standard Center): mean NPV \$592,516 (equal to the NPV at mean demand); loss probability 23.6% (236 of 1,000 trials); 5th-percentile NPV -\$760,364.

Limit: each trial holds one demand level for all three years, and demand is assumed normal.

WORKED EVIDENCE: STANDARD VERSUS FLEXIBLE CENTER PERFORMANCE

Practice case: 1,000 evenly spaced demand trials, $(i - 0.5)/1,000$, 3 years at 8%. Not your take-home options.

Design	Capital outlay	Mean NPV	NPV SD	P(NPV < 0)	5th-percentile NPV
Standard Center	\$2,500,000	\$592,516	\$824,547	23.6%	-\$760,364
Flexible Center	\$1,200,000	\$462,228	\$350,432	9.4%	-\$112,747

The Standard Center earns \$130,289 more on average but loses money in about one trial in four (23.6%), against about one in ten (9.4%) for the Flexible Center.

Demand mean 4,500, SD 800 seats, held for 3 years | Fixed overhead \$600,000 vs \$120,000 | Contribution \$400 vs \$170 per seat |
Unit: USD



*Lab 2 Guided
Implementation and
Sensitivity Analysis*

WORKED EVIDENCE: STANDARD VERSUS FLEXIBLE CENTER PERFORMANCE

Practice case, not your take-home options

1

Why the averages mislead

- The Standard Center earns \$130,289 more on average (\$592,516 against \$462,228) but loses money in 23.6% of trials, against 9.4%; its 5th-percentile NPV is $-\$760,364$ against $-\$112,747$.
- Its \$600,000 yearly fixed overhead (against \$120,000) magnifies every demand shortfall; the Flexible Center's low fixed cost works like insurance.

2

The mean alone cannot show the risk

- With a linear model the mean equals the deterministic value, so the risk shows only in the spread and the tail.
- Contrast Section 3: there the capacity cap made profit concave, so mean profit fell below profit at mean demand.

AI SEARCH & SHARE



*Discuss with a partner;
Artificial intelligence (AI)
support is optional.*

APPLY THE IDEA

Discuss with a partner; Artificial intelligence (AI) support is optional.

1

Partner question

- What is the operational interpretation of the 5th percentile Net Present Value metric in capital budgeting?
- Discuss with 1-2 neighbors. Prompt **AI**, check its answer, then volunteer one brief finding when ready.

DECISION POINT: IS THE HIGHER AVERAGE ALWAYS BETTER?

Use the evidence, explain your decision, and name one limit.



Decision question

The Standard Center earns more on average. Can the Flexible Center still be the better choice? Say when.

The evidence

Mean NPV: Standard \$592,516, Flexible \$462,228 (a \$130,289 gap).
Loss probability: 23.6% against 9.4%. 5th-percentile NPV: -\$760,364 against -\$112,747.

Name one limit

Practice case only: your take-home options use different inputs and will give different results.